R109/B_private_late | Method-conditioned nodes and actual latency

Complete standalone model including candidate node refinement; unequal parameter counts and no equal-FLOPs claim.

Problem	Instances	Milliseconds / instance
TSP	128	1.499769
CVRP	128	2.220458
ATSP	128	0.166184
OVRP	128	0.180281
VRPB	128	0.123409
VRPL	128	0.180113
VRPTW	128	0.180259
OVRPTW	128	0.180330
OVRPB	128	0.123008
OVRPL	128	0.179773
VRPBL	128	0.123186
VRPBTW	128	0.122917
VRPLTW	128	0.179700
OVRPBL	128	0.122899
OVRPBTW	128	0.122844
OVRPLTW	128	0.180226
VRPBLTW	128	0.123149
OVRPBLTW	128	0.122783

Total trainable parameters: 10,220,553; candidate-conditioned nodes, 21 global-ID-private two-layer graph banks

Optional R94B: not_requested

R109 refines each candidate node graph through two layers on shared original H4 before pooling. A_tied_late ties these weights; B_private_late has 21 banks keyed by global method ID, never candidate column, winner or outcome. Both retain candidate conditioning, original two joint layers, score MLP/maximin and original R77 CE/comparison/risk/two-forward RDropObjective. A has 2,305,033 trainables; B has 10,220,553. The original 1,842,953-parameter base is fresh matched, and all late branches start from identical original-layer prototypes. The initial eval function is identical, but complete state hashes/dimensions are not identical. This is not parameter/FLOP-matched, a routing experiment, or an isolated capacity-only control. Original R88B physical TSP node0 role, methodID+34 and geometry/distance remain; 57,600 updates, logical batch1024, seed2. No new route/path fields, labels, donor, teacher, context, targets, calibration, quotas or auxiliary loss. R89A AdamW, R89B Muon and R84A are saved historical references, not rerun or newly parameter/FLOP-matched controls; R94B is optional. NSS uses saved historical R79 weights, not retrained on new90k. Fixed36 true rescues/losses, false picks and aligned population net costs/regret are separate from selection frequency. Native/FP64 ind and legal pools; TRAIN-fitted SBS; ALL task-equal. TEST is development-exposed; one whole-model full18 VAL checkpoint per arm for full TRAIN/VAL/TEST/LIB; no TEST/LIB reselection. Actual standalone GPU-resident-input timing includes complete method-conditioned node refinement, excludes dataset I/O, CPU transfer and solvers, and is not end-to-end dataset latency. Microcaps/eval batch come from the main-witness locked runner, not this reporter. No routing summary is required. No online W&B equality/availability is asserted.

R109/B_private_late | Same selected model: full TRAIN

Diagnostic only; TRAIN was never used to select the checkpoint; full TRAIN-fitted SBS.

Problem	Top1 (%)	SBS Top1 (%)	mean_cost	SBS cost	Oracle cost	vs_SBS (%)	vs_Oracle (%)	actual regret (%)
ALL	71.9961	66.0332	11.387476	11.400702	11.345073	-0.2128	+0.3719	0.3793
TSP	47.1963	32.8617	8.004409	8.023959	7.970215	-0.2436	+0.4290	0.4257
CVRP	50.0160	18.1284	18.035509	18.151987	17.946816	-0.6417	+0.4942	0.6540
ATSP	76.9704	46.9210	1.639847	1.678679	1.633938	-2.3132	+0.3616	0.3634
OVRP	43.5025	34.2420	7.441533	7.453531	7.397676	-0.1610	+0.5928	0.6031
VRPB	81.4086	80.2765	9.352116	9.357264	9.319342	-0.0550	+0.3517	0.3570
VRPL	39.0679	31.5309	12.391256	12.400882	12.344503	-0.0776	+0.3787	0.3847
VRPTW	80.6543	80.2407	18.810817	18.813703	18.757954	-0.0153	+0.2818	0.2939
OVRPTW	84.0716	83.8025	10.117330	10.118640	10.093220	-0.0129	+0.2389	0.2331
OVRPB	88.6654	88.3580	6.362429	6.363230	6.346658	-0.0126	+0.2485	0.2547
OVRPL	43.6086	34.4716	7.442622	7.453219	7.398483	-0.1422	+0.5966	0.6057
VRPBL	80.5901	79.4593	9.398868	9.403115	9.363196	-0.0452	+0.3810	0.3813
VRPBTW	79.8963	79.4914	19.728181	19.733549	19.642712	-0.0272	+0.4351	0.4202
VRPLTW	80.6642	80.3827	18.767429	18.769959	18.714366	-0.0135	+0.2835	0.2970
OVRPBL	88.7877	88.5556	6.358103	6.358857	6.342513	-0.0119	+0.2458	0.2528
OVRPBTW	83.4506	83.2432	10.623035	10.624446	10.586225	-0.0133	+0.3477	0.3211
OVRPLTW	83.9642	83.7321	10.110020	10.111265	10.085768	-0.0123	+0.2405	0.2358
VRPBLTW	79.9247	79.5630	19.766827	19.770814	19.680364	-0.0202	+0.4393	0.4211
OVRPBLTW	83.4901	83.3370	10.624227	10.625544	10.587365	-0.0124	+0.3482	0.3229

ALL task-equal; ratios of mean costs differ from per-instance actual regret.

R109/B_private_late | Fixed target36 vs R89A

True rescues/losses, false picks and aligned net cost; fixed targets, no reselection.

Split	Winners	Rescued	Lost	New false	Net cost delta
TRAIN	83279	2074	49	1742	-0.001637105
VAL	4625	73	3	120	0.000226540
TEST	4598	71	6	121	0.000089644

Split	Top1 delta pp	Corrected	Harmed	Cost delta	Regret delta pp
TRAIN	0.751578	36527	25569	-0.001637105	-0.015410
VAL	0.124691	1754	1653	0.000226540	0.002682
TEST	0.128395	1744	1640	0.000089644	-0.000322

Historical only, not retrained/matched; R89A AdamW, R89B Muon, R84A AdamW.

Fixed-target-only deltas do not replace the full18 population net cost/regret above.

R109/B_private_late | Fixed target36 vs R89B

True rescues/losses, false picks and aligned net cost; fixed targets, no reselection.

Split	Winners	Rescued	Lost	New false	Net cost delta
TRAIN	83279	1534	202	1510	-0.000600779
VAL	4625	46	13	105	0.000764520
TEST	4598	48	12	102	0.000404678

Split	Top1 delta pp	Corrected	Harmed	Cost delta	Regret delta pp
TRAIN	0.304938	28882	24436	-0.000600779	-0.004343
VAL	-0.072840	1440	1499	0.000764520	0.007687
TEST	-0.059259	1454	1502	0.000404678	0.004268

Historical only, not retrained/matched; R89A AdamW, R89B Muon, R84A AdamW.

Fixed-target-only deltas do not replace the full18 population net cost/regret above.

R109/B_private_late | Fixed target36 vs R84A

True rescues/losses, false picks and aligned net cost; fixed targets, no reselection.

Split	Winners	Rescued	Lost	New false	Net cost delta
TRAIN	83279	2104	45	1757	-0.001684762
VAL	4625	77	3	119	0.000447327
TEST	4598	73	5	125	0.000032087

Split	Top1 delta pp	Corrected	Harmed	Cost delta	Regret delta pp
TRAIN	0.752195	35620	24653	-0.001684762	-0.015907
VAL	0.090123	1713	1640	0.000447327	0.004869
TEST	0.124691	1670	1569	0.000032087	-0.000933

Historical only, not retrained/matched; R89A AdamW, R89B Muon, R84A AdamW.

Fixed-target-only deltas do not replace the full18 population net cost/regret above.